

ENTERPRISE MILESTONE SPECIFICATION

Module 8: Hardware Integration, QR Handheld Devices, ESC/POS Printers & 3-Tier Server Setup

Physical computing infrastructure blueprint, peripheral driver daemon gateways, POS receipt & barcode printing pipelines, and on-premises high-availability 3-tier server virtualization for Popular Auto Spare & A/C Parts Co. W.L.L.

MODULE IDENTIFIER

SL-POP-ERP-MS-008

BA REFERENCE DOCUMENT

ARCH-001 v1.0 (04-July-2026)

TOTAL ENGINEERING EFFORT

3 Working Weeks (0.75 Months)

FIXED PROJECT COST

BD 1,142.727

TARGET ARCHITECTURE

On-Premise 3-Tier HA Cluster & Peripheral DAEMONS

DOCUMENT STATUS

FINAL PROPOSAL

Section 1.0

Executive Summary & Master ERP Context

Strategic Alignment

The **Hardware Integration & Infrastructure Platform** (Module 8) establishes the robust physical computing backbone, peripheral hardware driver gateways, thermal receipt printing pipelines, handheld Android PDA scanner synchronization daemons, and high-availability on-premises 3-tier server infrastructure for Popular Auto Spare & A/C Parts Co. W.L.L. It connects all physical retail counters, warehouse barcode terminals, label printers, and biometric attendance clocks directly to the centralized ERP core.



Core Objective & Business Purpose

To architect, deploy, and harden an enterprise-grade on-premises 3-tier virtualized server cluster, automated database replication, low-latency peripheral DAEMON gateways for Android PDA scanners, ESC/POS thermal receipt printers, Zebra industrial barcode printers, and multi-branch biometric time-clocks.

MASTER ERP TRANSFORMATION ARCHITECTURE (9 CORE PROCESS MODULES)

CURRENT SCOPE: MODULE 08
ACTIVE

01 PCode Gen & Item Master

02 Vendor & Purchase Flow

03 Stock Verification & Location

04 Sales & POS Checkout

05 Finance, Tax & VAT Accounting

06 Administration & Security

07 HRMS & Biometric Payroll

08 Hardware & Barcode Infrastructure

09 Executive BI & Mobile Analytics



3-Tier Virtualized Infrastructure

On-premise Linux cluster, Nginx reverse proxy, MariaDB master-slave replication, and daily automated SAN backups.

3-Tier HA

MariaDB Cluster

SAN Backups



Barcode & Handheld Integration

Android PDA scanner background daemons for real-time stock counting, goods receipt, and dispatch verification.

PDA Daemon

QR Scanners

Live Sync



POS Thermal & Label Printing

Direct network ESC/POS thermal receipt printers and Zebra ZPL industrial barcode shelf & item label generators.

ESC/POS

Zebra ZPL

Receipts

Section
2.0

Module 8: Functional Scope & Architecture Breakdown (BA Ref: ARCH-001)

Full Specification
Mapping

Derived directly from **ARCH-001: Technical Architecture & Hardware Infrastructure Blueprint**, the implementation scope is structured across 5 core technical pillars:

2.1 On-Premises 3-Tier Server Virtualization & Database Clustering

- **Proxmox VE / KVM Virtualization:** Deployment of isolated Virtual Machines for Web/App layer, Database layer, and Background DAEMON services.
- **Database High Availability:** MariaDB replication cluster with automated failover and daily point-in-time recovery (PITR) snapshots.
- **Nginx Reverse Proxy & Load Balancer:** SSL termination, HTTP/2 compression, request throttling, and DDoS protection filters.
- **Automated Local & Cloud Backup Pipeline:** AES-256 encrypted night-time database dumps and transaction log sync to remote off-site storage.

2.2 Android PDA Handheld Barcode / QR Scanner Integration (DAEMONS)

- **Handheld Hardware Compatibility:** Integration with Honeywell, Zebra, and Urovo industrial Android rugged PDA barcode scanners.
- **Low-Latency REST/WebSocket DAEMON:** Sub-second verification of scanned PCodes, bin locations, and purchase order items.
- **Offline Buffer Mode:** Local SQLite caching on handhelds ensuring warehouse picking and stock taking proceed during Wi-Fi blind spots.
- **Handheld Device Management:** Device authorization tokens, remote configuration push, and scanner battery telemetry monitoring.

2.3 ESC/POS POS Thermal Receipt & Zebra Industrial Barcode Label Printing

- **Direct Network ESC/POS Driver:** Raw socket printing to Epson/Bixolon 80mm POS thermal receipt printers bypassing browser print dialogs.
- **Zebra Programming Language (ZPL) Engine:** High-speed dynamic generation of 2D QR and Code-128 barcode shelf tags and item labels.
- **Multi-Terminal Print Spooler:** Centralized queue management handling concurrent receipt prints across all checkout counters without collision.
- **VAT-Compliant Invoice Layouts:** Dynamic printing of NBR-compliant simplified VAT tax invoice receipts with embedded cryptographic QR codes.

2.4 Multi-Branch Biometric Attendance Device Gateway (ZKTeco Protocol)

- **ZKTeco Standalone Push Protocol:** Gateway service capturing live biometric clock-in / clock-out punches from branch attendance terminals.
- **Device User Synchronization:** Automated two-way enrollment push of staff biometric IDs, names, and card numbers to all branch terminals.
- **Heartbeat Health Monitor:** Real-time status ping tracking online/offline connectivity of biometric clocks across all retail branches.

2.5 Network Security, SSL/TLS Encryption & Automated SAN Backup Vault

- **End-to-End SSL/TLS Encryption:** HTTPS certificates deployed across all local intranet endpoints and mobile device endpoints.
- **VLAN Network Segmentation:** Segregation of POS terminals, server cluster, Wi-Fi handhelds, and guest networks via managed switches.
- **Automated SAN Backup Vault:** Nightly differential and weekly full backup archives with automatic 30-day retention pruning.

Section 3.0

Project Team Structure & Dedicated Resource Pricing (3 Weeks / 0.75 Months)

Standard
Rate Card

In strict accordance with the **SaNDS Lab Dedicated Engineering Rate Card**, the resource allocation and cost breakdown for the 3-week (0.75 months / 15 working days) delivery of Module 8 is structured as follows (all figures in Bahraini Dinars - BHD):

ENGINEERING ROLE	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H/MO)	EFFORT ALLOCATION	TOTAL COST (BHD)
Project Manager & Enterprise Solutions Architect	BD 545.455	BD 136.364	BD 3.409	0.75 Months (3 Wks)	BD 409.092
Back-End Lead Engineer (Hardware Drivers / API Gateway)	BD 227.273	BD 56.818	BD 1.420	0.75 Months (3 Wks)	BD 170.454
Front-End Lead Engineer (Device Management Console)	BD 227.273	BD 56.818	BD 1.420	0.75 Months (3 Wks)	BD 170.454
Database & Cloud Infrastructure Architect	BD 204.545	BD 51.136	BD 1.278	0.75 Months (3 Wks)	BD 153.408
Hardware Systems & Network Security Specialist	BD 318.182	BD 79.546	BD 1.989	0.75 Months (3 Wks)	BD 238.638
Total Dedicated Engineering Team	BD 1,523.636 / mo	BD 380.909 / wk	BD 9.518 / hr	3 Weeks (0.75 Mo)	BD 1,142.727



Fixed Price Guarantee

Total project investment for Module 8 is fixed at **BD 1,142.727**. No hidden deployment, integration, or hourly surcharges apply within the agreed functional scope.

The implementation is organized into 3 sequential 1-week milestones (33.33% / 33.34% each), guaranteeing rapid infrastructure deployment and verified peripheral integration:

1

Milestone 8.1: 3-Tier Server Virtualization, OS Hardening & Database Clustering

Week 1 (1
Wk)

Setup of on-premises Proxmox/KVM hypervisor, Linux VM provisioning, MariaDB high-availability master-slave replication, Nginx SSL reverse proxy, and local automated daily backup cron tasks.

- ✓ **Server Hypervisor & VMs:** Provisioning of dedicated Web/App, DB, and DAEMON virtual machines with resource limits.
- ✓ **Database HA Cluster:** Replication configuration, automated failover scripts, and daily backup dumps.
- ✓ **SSL & Security Hardening:** UFW firewall configuration, SSH key policies, and Nginx SSL certificates.

2

Milestone 8.2: Android PDA Scanner Daemon & ESC/POS / Zebra Label Printing Engine

Week 2 (1
Wk)

Deployment of Android PDA barcode scanning backend daemon with offline buffer, raw ESC/POS network thermal receipt printing driver, and Zebra ZPL industrial label generation pipeline.

- ✓ **PDA Scanner Service:** Sub-second PCode scanning verification and warehouse picking synchronization.
- ✓ **ESC/POS Receipt Driver:** Direct network thermal printing of simplified VAT tax invoice receipts.
- ✓ **Zebra ZPL Label Engine:** Dynamic QR and Code-128 barcode shelf and part label generation.

3

Milestone 8.3: Biometric Device Gateway, Automated Backup Vault & Go-Live Cutover

Week 3 (1 Wk)

Setup of ZKTeco biometric attendance gateway service, automated SAN backup verification drill, end-to-end peripheral stress testing across all retail counters, and final production cutover sign-off.

- ✓ **Biometric Device Gateway:** Live punch sync daemon from multi-branch ZKTeco clocks into HRMS.
- ✓ **Backup Restoration Drill:** Verification of point-in-time recovery and automated SAN backup replication.
- ✓ **Hardware UAT Sign-Off:** Full counter printer, scanner, and server verification with stakeholder acceptance.

Section 5.0

Milestone Payment Structure & Invoicing Schedule (3-Week Model)

Commercial Billing Governance

Billing is structured across 3 sequential verification gates tied directly to verifiable deliverable sign-offs:

GATE	MILESTONE DELIVERABLE FOCUS	TIMELINE	% FEE	INVOICE AMOUNT (BD)	VERIFICATION CRITERIA
Gate 8.1	3-Tier Server Virtualization & Database Clustering	End of Wk 1	33.33%	BD 380.909	Dev/Staging/Prod instances online & SSL verified
Gate 8.2	PDA Scanner Daemon & ESC/POS / Zebra Label Printing	End of Wk 2	33.33%	BD 380.909	PDA PCode barcode scan & receipt print verified
Gate 8.3	Biometric Clock Sync, SAN Backup & Final Go-Live	End of Wk 3	33.34%	BD 380.909	Biometric punch push verified & backup drill approved
Total Module 8 Investment (Fixed Deliverable Contract)			100.00%	BD 1,142.727	Complete Production Delivery

Section 6.0

Working Calendar, SLA, 15-Day Grace Period & Legal Terms

Master Service Level Agreement

This milestone schedule is governed strictly under the overarching Master Payment Governance & Service Level Framework (SL-POP-ERP-PAY-001):

 Standard Working Calendar & Effort Accounting

All engineering effort calculations are structured upon a standard 5-day working week (Sunday through Thursday), 8 working hours per day (40 hours/week, 160 hours/month). Kingdom of Bahrain statutory public holidays are observed.

15-Day Verification Window & Deemed Acceptance

Upon milestone deployment notification, the Client is granted a 15–calendar–day UAT verification period. If no material defect is logged in writing within 15 days, milestone acceptance and invoice authorization occur automatically.

90-Day Production Bug-Fix Warranty

A comprehensive 90–calendar–day post–go–live warranty covers all functional discrepancies, calculation anomalies, and system defects originating from the agreed ARCH-001 specification without additional charge.

2-Year Resource Rate Protection Guarantee

Engineering resource rates defined in Section 3.0 remain legally fixed for twenty-four (24) months from contract execution, ensuring predictable costs for any post–implementation change requests.

Section
7.0

Stakeholder Authorization & Digital Sign-Off Console

Cryptographic Audit
Trail

By executing digital signatures below, all authorized representatives formally validate the functional architecture, 3–week delivery roadmap, resource allocation pricing, and invoicing governance defined in this document:

FIRST PARTY – CLIENT / END–USER

Popular Auto Spare & A/C Parts Co. W.L.L.

Awaiting Client Signature

Signatory: Authorized Executive Director

Status: Pending Digital Execution

SECOND PARTY – LEAD IMPLEMENTATION PARTNER

SaNDS Lab Software Solutions

Awaiting SaNDS Lab Signature

Signatory: Lead Solutions Architect

Status: Pending Digital Execution

THIRD PARTY – ADVISORY & ASSURANCE PARTNER

UniGlobal Business Solutions

Awaiting UniGlobal Signature

Signatory: Principal Consultant

Status: Pending Digital Execution